

Objective

Build a system that consumes a complex Port Tariff PDF and a natural language vessel query to automate the calculation of different port dues that may be payable by a vessel arriving at a port.

The Core Challenge

Port tariff documents are dense, hierarchical, condition-based, often containing a lot of tabular data. We are looking for a **generalizable solution** that can tell us what charges/tariffs are applicable for a vessel arriving at a port according to the port's tariff document(s).

Ideally your solution is an agentic system, but this is not a requirement – you are free to build whatever AI system you feel is most suitable for this task.

Crucially: Your architecture should be robust enough to handle similar tariff structures from other ports or future document updates with minimal manual code changes. We want to see how your agent "finds" and "interprets" the logic from a typical tariff document, to be able to automate calculating the tariffs without manual intervention and hardcoded information.

Required Outputs

For a given vessel and port (e.g., Durban, Amsterdam, etc), your solution must calculate all the applicable tariffs (like, light dues, port dues, towage dues, VTS dues, pilotage dues, running of vessel lines dues, etc)

Reference Data for Validation

The solution you build must be general for any vessel at any port. To test it, you can use the attached South African tariff document and the following vessel profile for the port of Durban to verify your calculations against our ground truth values:

- **Vessel Details:**

```
{  
  "vessel_metadata": {  
    "name": "SUDESTADA",  
    "built_year": 2010,  
    "flag": "MLT - Malta",
```

```
"classification_society": "Registro Italiano Navale",  
"call_sign": null  
},  
"technical_specs": {  
  "imo_number": null,  
  "type": "Bulk Carrier",  
  "dwt": 93274,  
  "gross_tonnage": 51300,  
  "net_tonnage": 31192,  
  "loa_meters": 229.2,  
  "beam_meters": 38.0,  
  "moulded_depth_meters": 20.7,  
  "lbp_meters": 222.0,  
  "draft_sw_s_w_t": [14.9, 0.0, 0.0],  
  "suez_gt": null,  
  "suez_nt": 49069  
},  
"operational_data": {  
  "cargo_quantity_mt": 40000,  
  "days_alongside": 3.39,  
  "arrival_time": "2024-11-15T10:12:00",  
  "departure_time": "2024-11-22T13:00:00",  
  "activity": "Exporting Iron Ore",
```

```

    "num_operations": 2,

    "num_holds": 7

  }

}

```

- **Reference Outputs**

Tariff Item	Expected Value (Approx. ZAR)
Light Dues	60,062.04
Port Dues	199,549.22
Towage Dues	147,074.38
VTs Dues	33,315.75
Pilotage Dues	47,189.94
Running Lines	19,639.50

Technical Requirements

- **Stack:** Any framework (LangChain, LlamaIndex, etc.). We recommend using **Gemini LLMs** (because of their [free tier with generous rate limit](#)) for extraction and reasoning. You can use anything like Claude Code and any setup you find workable for you.
- **Deliverable:** A GitHub repository containing:
 - Code or md files.
 - A README.md with clear setup and execution instructions.
 - **Bonus:** Deploy the solution as an API endpoint.

Evaluation Criteria

1. **Accuracy of tariff calculations:** How close are your calculations to the ground truth?
2. **Level of Automation:** Does the agent "find" the rules itself, or is the logic hard-coded? As mentioned, the solution needs to be a general one that can accommodate any port and tariff document.
3. **Code Quality:** In your scenario it just has to work.

Questions? Please email bd@marcura.com directly for any clarifications.